

Zekun Sun
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Academic Appointment

2023– Postdoctoral Associate, [Action, Computation and Thinking Lab](#), Yale University (Advisor: [Samuel D. McDougle](#))

Education

2017–23 **Ph.D.**, Cognitive Psychology, Johns Hopkins University
(Advisor: [Chaz Firestone](#))

2013–16 **M.S.**, Cognitive Neuroscience, Chinese Academy of Sciences

2009–13 **LL.B.**, International Politics, Nanjing University

Publications

 [Google Scholar](#)

Journal Articles

- J1. **Sun, Z.** & McDougle, S. D. Perceiving event structure in brief actions. *Cognitive Psychology* (in press).
- J2. **Sun, Z.**, Firestone, C. & Hafri, A. The psychophysics of compositionality: Relational scene perception occurs in a canonical order. *Cognitive Psychology* **161**, 101765 (2025).
- J3. **Sun, Z.** & Firestone, C. Caricaturing shapes in visual memory. *Psychological Science* **35**, 722–735 (2024).
- J4. **Sun, Z.** & Firestone, C. Beautiful on the inside: Aesthetic preferences and the skeletal complexity of shapes. *Perception* **51**, 904–918 (2022).
- J5. **Sun, Z.** & Firestone, C. Seeing and speaking: How verbal “description length” encodes visual complexity. *Journal of Experimental Psychology: General* **151**, 82 (2022).
- J6. **Sun, Z.** & Firestone, C. Curious objects: How visual complexity guides attention and engagement. *Cognitive science* **45**, e12933 (2021).
– **Top-cited article in *Cognitive Science*, 2022.**

- J7. **Sun, Z.** & Firestone, C. Optimism and pessimism in the predictive brain. *Trends in Cognitive Sciences* **24**, 683–685 (2020).
- J8. **Sun, Z.** & Firestone, C. The dark room problem. *Trends in cognitive sciences* **24**, 346–348 (2020).
- J9. Fan, L., Sun, Y.-B., **Sun, Z.**, Wang, N., Luo, F., Yu, F. & Wang, J.-Y. Modulation of auditory sensory memory by chronic clinical pain and acute experimental pain: a mismatch negativity study. *Scientific Reports* **8**, 15673 (2018).
- J10. **Sun, Z.**, Wang, J.-Y. & Luo, F. Experimental pain induces attentional bias that is modified by enhanced motivation: An eye tracking study. *European Journal of Pain* **20**, 1266–1277 (2016).

Working papers

- W1. **Sun, Z.**, Bai, D., Scholl, B. J. & McDougle, S. D. *Visuomotor adaptation via featureless objects.* ([full manuscript](#)).
- W2. **Sun, Z.** & McDougle, S. D. *Motor planning is obligatorily biased by task-irrelevant objects.* ([full manuscript](#)).
- W3. **Sun, Z.**, Xie, Z. & McDougle, S. D. *Novel skills recruit motor abstractions.*

Presentations

Invited Talks

- I1. Perception and Action Seminar, Brown University. Sept. 2025.
Title: Representational abstraction in visual and motor cognition.
- I2. The Skilled Action and Memory Lab, University of Birmingham. July 2025.
Title: Novel skills recruit motor abstractions.
- I3. Cognitive Current Works, Yale University. Feb. 2025.
Title: Novel skills recruit motor abstractions.
- I4. The Action, Cognition and Thinking Lab, Yale University. June 2022.
Title: Cognitive consequences of visual complexity.
- I5. Behrmann Lab, Carnegie Mellon University. Apr. 2022.
Title: Cognitive consequences of visual complexity.
- I6. The Computational Vision and Learning Lab, University of California, Los Angeles. Apr. 2022.
Title: Cognitive consequences of visual complexity.
- I7. The Language and Cognition Lab, Stanford University. Jan. 2022.
Title: Cognitive consequences of visual complexity.

- I8. Cognitive Tools Lab, University of California, San Diego. Oct. 2021.
Title: Cognitive consequences of visual complexity.
- I9. The Computation and Language Lab, University of California, Berkeley.
Oct. 2021.
Title: Cognitive consequences of visual complexity.

Talks

- T1. **Sun, Z.** & D., M. S. *Dissociating external features from internal structures in visual segmentation of actions.* The 25th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2025.
- T2. Hafri, A., **Sun, Z.** & Firestone, C. *The psychophysics of compositionality: Relational scene perception occurs in a canonical order.* The 24th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- T3. **Sun, Z.**, Bi, W., Yildirim, I. & McDougale, S. D. *Breaking down a golf swing: Spatio-temporal dynamics of visual motion underlie high-level structuring of observed actions.* The 24th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- T4. **Sun, Z.**, Han, S. & Firestone, C. *The evolution of complexity in visual memory.* The 21st annual meeting of the Vision Sciences Society (online). Nov. 2021.
- T5. Yu, Q. & **Sun, Z.** *Looking unique.* the 29th annual meeting for Object Perception, Attention, and Memory (online). Nov. 2021.
- T6. **Sun, Z.** & Firestone, C. *Speaking about seeing: How verbal descriptions encode visual complexity.* The 27th annual meeting of Object Perception, Attention, and Memory (Montreal, Canada). Nov. 2019.
– **Best Talk Award.**
- T7. **Sun, Z.** & Firestone, C. *Curious objects: Preattentive processing of object complexity.* The 18th annual meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). Nov. 2018.

Posters

- P1. **Sun, Z.**, Bai, D., Scholl, B. J. & McDougale, S. D. *Motor Adaptation via Pure Visual Change.* Neural Control of Movement 2025 (Panama City, Panama). July 2025.
- P2. **Sun, Z.** & McDougale, S. D. *Novel motor skills recruit visuomotor abstractions.* The 25th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2025.

- P3. Kang, S., **Sun, Z.** & McDougale, S. D. *Investigating Structural Learning in a De Novo Motor Skill.* Society for Neuroscience 2024 (Chicago, IL, USA). Oct. 2024.
- P4. **Sun, Z.**, Hafri, A. & Firestone, C. *How to build a scene: Relational representations are constructed in a canonical order.* The 24th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- P5. **Sun, Z.** & McDougale, S. D. *Goal uncertainty biases memory for observed actions.* The 24th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- P6. **Sun, Z.** & McDougale, S. D. *Repelling Effects of Distractors Persist in Imitated Movements.* Society for Neuroscience 2024 (Chicago, IL, USA). Oct. 2024.
- P7. Han, S., **Sun, Z.** & Firestone, C. *Caricaturing shape in visual memory.* The 23rd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2023.
- P8. Hafri, A., **Sun, Z.** & Firestone, C. *Sequential construction of visual relations.* The 22nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2022.
- P9. **Sun, Z.**, Yu, Q. & Firestone, C. *How to look unique.* The 22nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2022.
- P10. **Sun, Z.** & Firestone, C. *The evolution of complexity in visual memory.* Object Perception, Attention, and Memory 2021 (Online). Nov. 2021.
- P11. Halberda, J., Yu, Q., **Sun, Z.** & Firestone, C. *Not too simple, not too complex: The Goldilocks principle drives discrimination and search.* The 20th Annual Meeting of the Vision Sciences Society (Online). June 2020.
- P12. **Sun, Z.** & Firestone, C. *The simple and the beautiful: An aesthetic preference for medially complex stimuli.* The 20th Annual Meeting of the Vision Sciences Society (Online). June 2020.
- P13. **Sun, Z.** & Firestone, C. *Speaking about seeing: How verbal descriptions encode visual complexity.* The 19th Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2019.
– **Student Travel Award.**

Awards & Honors

2022	G. Stanley Hall's Scholar Award Johns Hopkins University
2021	Clark Collaborative Award Johns Hopkins University
2021	The Mary Ainsworth Award Johns Hopkins University
2019	The Robert S. Waldrop Junior Investigator's Award Johns Hopkins University
2019	OPAM Best Talk Award Object Perception, Attention, and Memory
2019	Elsevier/Vision Research Travel Award Vision Sciences Society
2018/2019	Collaborative Research Award Johns Hopkins University
2015/2016	Graduate Student Award Chinese Academy of Sciences

Teaching

Johns Hopkins University

2020	Teaching Assistant, Real World Human Data
2019	Teaching Assistant, Methods in Experimental Psychology
2019	Teaching Assistant, Introduction to Cognitive Psychology
2018	Teaching Assistant, Introduction to Social Psychology

Academic Advising

Graduate

2025 Zhiran Xie (Master student in Computational Biology), Yale University. Research project on motor control and neural networks, research assistant

Undergraduate

2025 Emma Kohlmayer (Art History and Cognitive Science), Yale University. Research project on action and perception and senior thesis.

2024 Zack Haaland (Cognitive Science), Yale University. Senior thesis.

2022/2023 Subin Han, Johns Hopkins University. Psychology project internship that resulted in Hopkins undergraduate research award (PURA) and a co-authored publication in *Psychological Science*.

Academic Service

Intern Junior Editor

2025 – 2026 Journal of Experimental Psychology:
Human Perception and Performance

Journal Reviewer

Attention, Perception, and Psychophysics

iPerception

iScience

Journal of Cognitive Neuroscience

Journal of Experimental Psychology: Human Perception and Performance

Journal of Experimental Psychology: General

Philosophical Transactions of the Royal Society B

Perception

Synthese

Visual Cognition

Other experience

- 2020 Early Career Colloquium Selection Committee,
 — Johns Hopkins University
 Role: Student Organizer
- 2019 Summer Internship Program
 — Johns Hopkins University
 Role: Mentor, Vision Group
 Project: Shape bias and complexity bias
 Mentee: Subin Han, Cognitive Science Department
- 2019 Department Colloquium Committee
 — Johns Hopkins University
 Role: Student Organizer

References

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Last updated: October 13, 2025